

Permission to record Dhaka roads

For a university research dataset. A marked vehicle, driven normally. No person is approached, and no number plate is ever read.

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The vehicle, marked and openly equipped

Mitsubishi L200 • sensors visible on the roof frame

— WHAT WE ARE ASKING

Permission to drive and record on public roads

One university vehicle. Two crew. Normal, legal driving. Cameras and sensors facing outward at the road.

1

vehicle

Marked with the BRAC University name, equipment mounted openly.

2

crew

A dedicated driver and a separate recording operator.

15→200

hours

At least 15 hours in the first phase, then extended collection.

4

road types

Commercial, arterial, older narrow-street and residential.

WHAT WE ARE NOT ASKING FOR

No road closure, no escort, no traffic management, no access to any restricted area, and no data from the police. We ask only to be permitted to drive in ordinary traffic with recording equipment visible on the vehicle.

— THE PROBLEM

A ground robot must answer two questions, continuously

01

Where am I?

A robot finds its position from satellite signals — which fail in exactly the places it has to work.

- Between tall buildings, under flyovers, in narrow lanes, near large metal structures.
- Signals bounce off surfaces before arriving, so the position comes out confident and wrong.

A machine that trusts it will drive into a wall.

02

Where can I go next?

A route that looks clear on a map is often not one a machine can physically cross.

- A broken slab, an open drain, a spoil heap from repair work, a kerb too high to climb.
- The ground has to be read as shape and surface — along with everything moving across it.

Neither question can be answered from a computer simulation.

— THE SOLUTION

Record the real road, then measure against it

No dataset of this kind exists for Dhaka. The vehicle stands in for the robot: it carries the same sensors, and is driven normally through the traffic a robot would one day have to handle.

01

Record

Drive ordinary routes in ordinary traffic while outward-facing sensors record the road.

02

Anonymise

Every face and every number plate is automatically blurred. Speech is removed from the sound.

03

Develop

Use the recordings to teach a machine to find its own position and judge where it can drive.

04

Publish

Release the anonymised result for non-commercial research use, so other universities can build on it.

WHY DHAKA

Dhaka provides in ordinary conditions what other cities can only create artificially: satellite reception that genuinely degrades between buildings and under flyovers, irregular road surfaces, and traffic that negotiates right of way rather than following lanes. A method that works here has been tested against a harder case than an orderly city can offer.

EXISTING WORK

Camera datasets exist across South Asia. Navigation datasets do not.

Four countries have built public road datasets; only India has gone past the camera — with a rig much like ours. What nobody has built is the Bangladeshi one.



Bangladesh — BadODD: nine districts, day and night, 2D boxes on smartphone photographs.

Baig, Hajong, Patwary, Rahman, Chowdhury (2024), arXiv:2401.10659 • CC BY 4.0



India — the IDD-3D collection car: six cameras and a roof LiDAR. The same configuration we propose, on a smaller vehicle.

Dokania et al., WACV 2023, arXiv:2210.12878 • CC BY 4.0



What that rig produces: every road user boxed in three dimensions and tracked, with the camera views alongside.

Dokania et al., WACV 2023, arXiv:2210.12878 • CC BY 4.0

India

IDD • IDD-3D • DriveIndia

LiDAR and camera, 3D boxes

Pakistan

CARL-D • R2S100K • PDrive20K

Stereo camera, road-region masks

Bangladesh

BadODD • BNVD • BRSSD10K • Poribohon-BD

Single camera, 2D boxes and masks

Sri Lanka

CeyRo

Traffic signs and lights only



The equipment is mounted openly and is visible from outside. The vehicle carries the university name.
BRAC University, Department of Computer Science & Engineering

— THE VEHICLE

One marked pickup, sensors facing outward

Cameras, front and rear — ordinary colour video of the road ahead and behind

Depth cameras, each side — measure the shape of the ground and how far away things are

Laser scanner on the roof — measures the shape of the surroundings, like a radar

Satellite antennas and motion unit — record the path of our own vehicle only

Microphone — ambient traffic sound, so a vehicle out of sight can be heard

The equipment is low in profile and adds negligible weight. The vehicle remains road legal, handles normally, and is insured for this use.

— RECORDING

What is recorded — and what never is

RECORDED

- ✓ Video of the road ahead, behind and to the sides
- ✓ The shape of the road surface and of nearby objects
- ✓ The path of our own vehicle
- ✓ General traffic sound

NEVER RECORDED

- ✗ No number plate is read, written down or stored as text — at any stage
- ✗ No conversation. Speech is detected and removed from the sound
- ✗ No recording inside any other vehicle
- ✗ No camera zooms in and no microphone is pointed at any person
- ✗ No name, address or personal detail about anyone
- ✗ No attempt to identify anyone, or to link a recording to any other record

— CONDUCT ON THE ROAD

How the vehicle behaves in traffic

Driving is normal and legal

No manoeuvre is ever performed for the benefit of the sensors. A dedicated driver drives and does nothing else; a separate operator runs the recording and does not drive.

Papers are carried in the vehicle

A departmental authorisation letter and a short plain-language explanation of the project are in the vehicle at all times, along with proof of insurance.

Nobody is followed or waited for

The vehicle does not follow, approach or wait on any individual, and does not stay at any location longer than traffic requires.

Sensitive places are avoided

Recording near schools, hospitals, places of worship and public gatherings is avoided. Anything caught inadvertently is deleted rather than published.

We stop when asked

The crew answer questions from the public, stop recording if anyone objects, and comply immediately with any instruction from a police officer.

Sessions end early when conditions change

Recording is abandoned when rain begins, when a crowd gathers around the vehicle, or in any situation that would require driving other than normally.

— PRIVACY

Nothing identifiable is published

01 Automatic blurring

Faces and number plates, across the whole release — not a sample.

02 Checked by a person

One miss sends the whole batch back for reprocessing.

03 Speech removed

Speech is detected and deleted. Only traffic noise is published.

04 Nothing leaves the university

No cloud service, no outside company.



Every face is detected and blurred before anything leaves the university. Number plates are treated the same way.

Illustration produced from our own photograph, BRAC University

ANYONE CAN ASK FOR REMOVAL

A published contact point accepts removal requests from anyone, and honours them without asking for a justification.

— IF SOMETHING HAPPENS

Arranged responses, agreed in advance

If a collision or injury is recorded

The crew stop recording and help — call emergency services and assist as any road user would. The session is logged as ended and that footage is excluded from the public release.

If a lawful order is made

The project would comply with an order from a court or competent authority. This limit is written into the consent form our own crew sign, so nothing is promised that could not be honoured.

If a traffic offence is caught on camera

The project does not review footage for offences, does not report it to any authority, and does not use it to identify anyone. No number plate is ever read, so no vehicle in this footage can be traced to an owner through our dataset.

If anyone objects to being recorded

The crew stop. A session ended by request is logged as such. Any complaint, incident or lost storage device is reported to the Principal Investigator within one working day.

— WHERE AND WHEN

Four road types, by day and at night

Routes are chosen by road type, never by anything about the people using them. Specific corridors to be agreed with DMP.

Dense commercial streets

Arterial roads

Older narrow-street areas

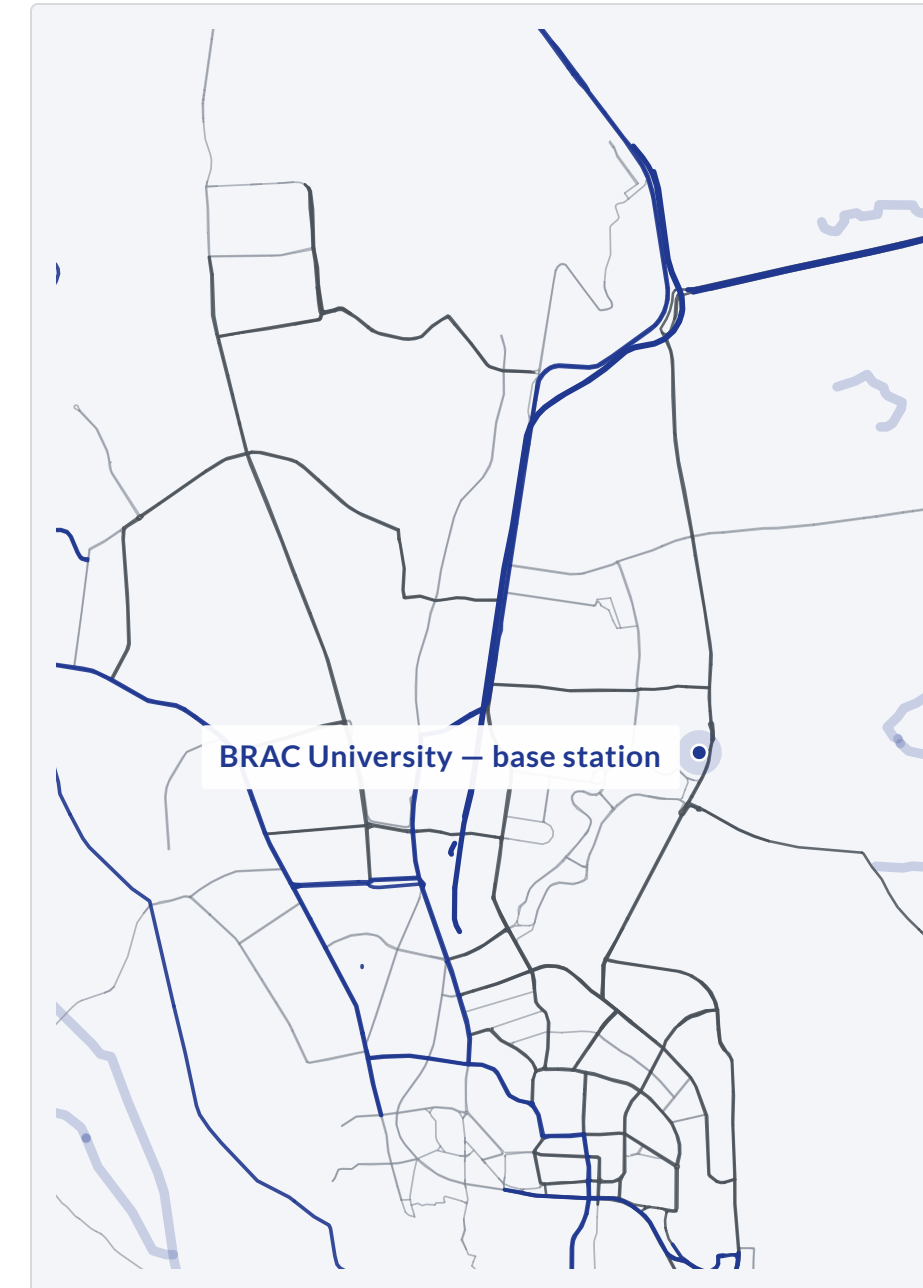
Residential streets

15 hrs

Minimum in the first phase, after two short pilot sessions.

200 hrs

Target for extended collection, as storage allows.



— Motorway & trunk — Primary — Secondary

Map data © OpenStreetMap contributors (ODbL)

— APPROVALS AND LEGAL BASIS

What is already in place

University ethics approval

The study is under review by the Institutional Review Board of BRAC University. No recording takes place until that approval is in place.

Personal Data Protection Ordinance, 2025

The Ordinance exempts processing for research and statistics where the data subject is not identified as a result. This study is designed to sit inside that exemption, not to rely on it after the fact: no individual is identified at any stage.

Departmental authorisation

A written statement of support and vehicle authorisation letter from the Department of Computer Science and Engineering is carried in the vehicle.

No external sponsor

Funded from BRAC University internal resources. No company is involved in the design, collection or release, and the licence permits non-commercial research use only.

— OUR REQUEST

What we ask of the Dhaka Metropolitan Police

We record the road, not the people on it.

1

Written permission to operate the marked research vehicle on public roads in Dhaka, day and night.

2

A point of contact within DMP we can notify before each collection phase.

3

Guidance on any road, area or time we should avoid.

CONTACT

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Figure credits: BadODD and IDD-3D figures reproduced under CC BY 4.0 with attribution. Full detail: the complete data collection plan, storage tiers, labelling procedure, consent forms and release arrangements are set out in the IRB application submitted to BRAC University.

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